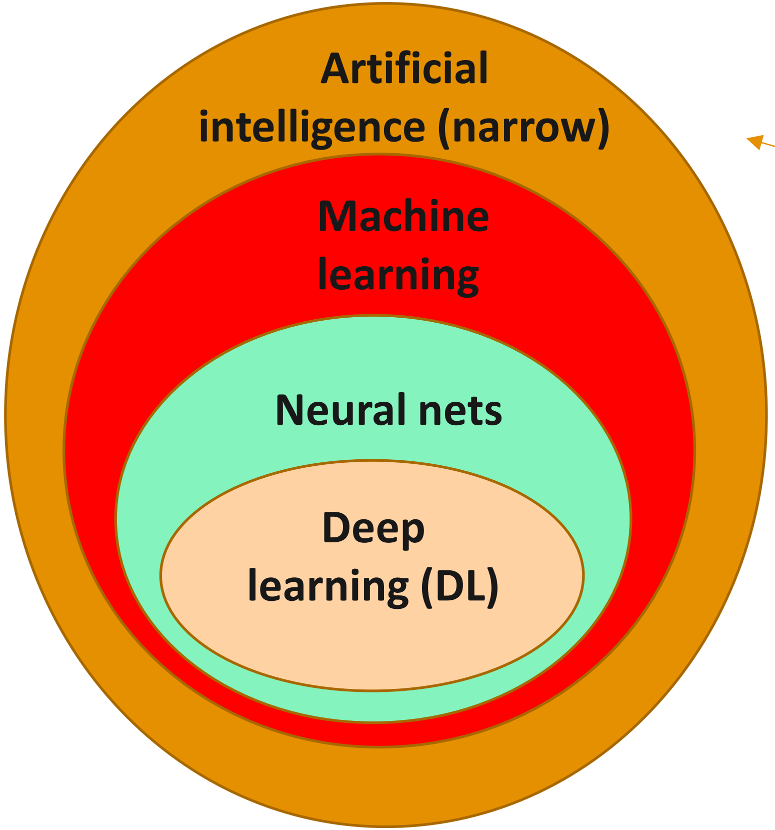
Machine Learning Paradigms

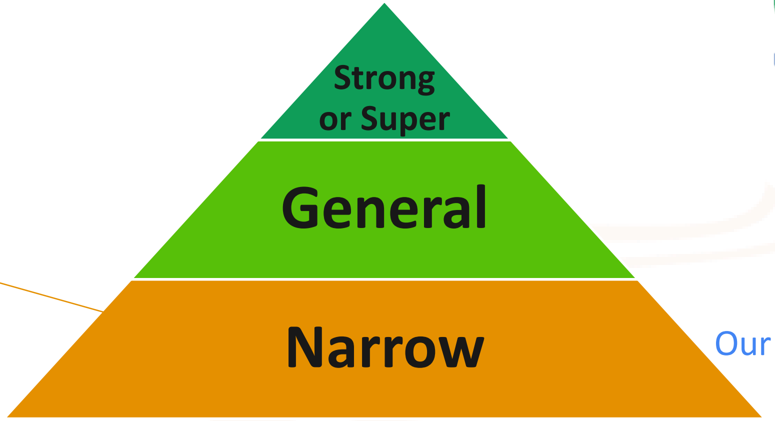
Aldebaro Klautau
Federal University of Pará (UFPA) / LASSE

Computational Intelligence Class

Aug. 07, 2026

AI, Machine Learning and Deep learning (DL)

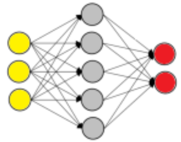




Our scope

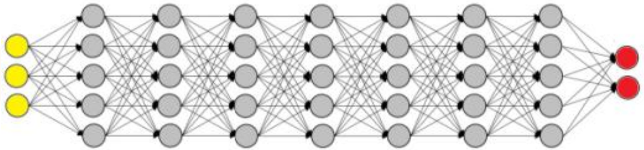
DL is a set of techniques to **train** and **deploy** neural networks with large number parameters

Shallow network **Input**



Output

Deep network



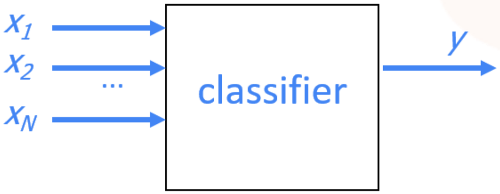
Os dois principais paradigmas que estudaremos:

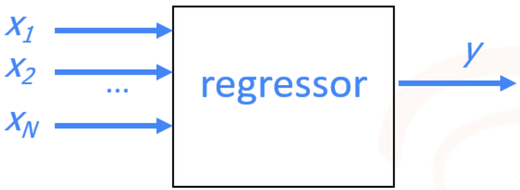
Classificação

Exemplos de y: 0, 1 ou 2

Regressão

Exemplos de y: -5.4 ou 3.1





GÉRON, Aurélien. **Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems**. " O'Reilly Media, Inc.", 2019.

Simple classifiers (for two simple sets)

- Let us assume a binary classification problem and the two simple sets below:

Training set

Length	Weight	Class y
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

Test set

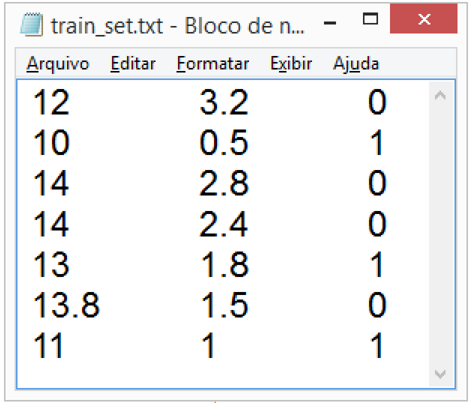
Length	Weight	Class y
13	3.1	0
9	0.8	0
12.3	1.4	1
10	2.3	1

Getting familiar with the given data

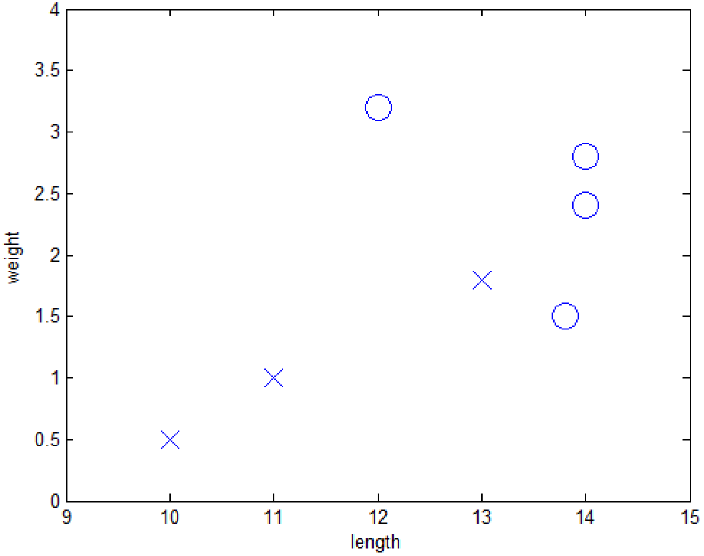
- Because there are (only) two features, it is easy to visualize the training set

Training set

Length	Weight	Class y
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1



Text (ASCII) file



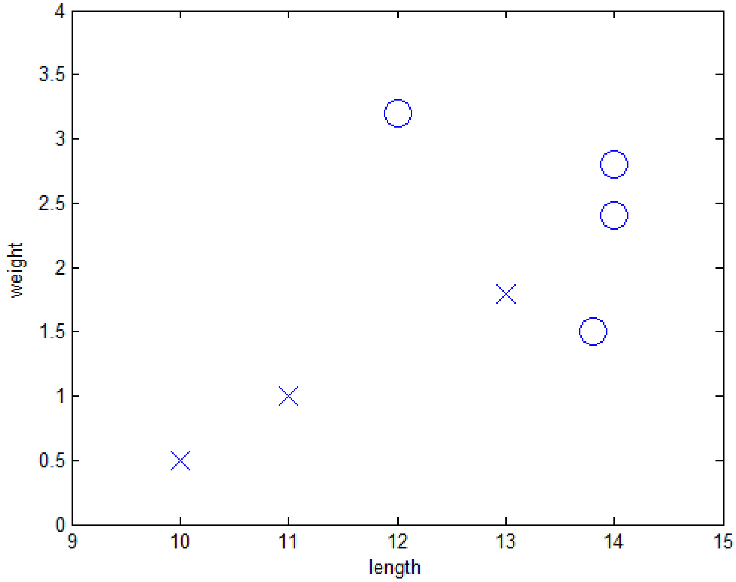
First classifier: decision stump

train_set.txt - Bloco de n...

Arquivo Editar Formatar Exibir Ajuda

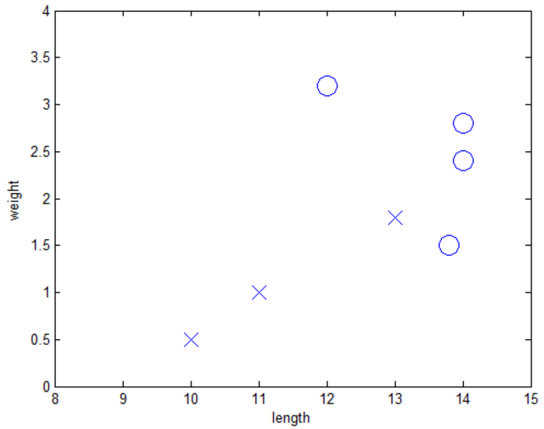
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

- Decision stump is a single if / else rule based on a chosen threshold value of a chosen feature
- First example:
  - if weight > 1
    - then class label is 0
  - This gives one error in training set!
- Another example:
  - if length < 12
    - then class label is 1
  - This also gives one error in training set!

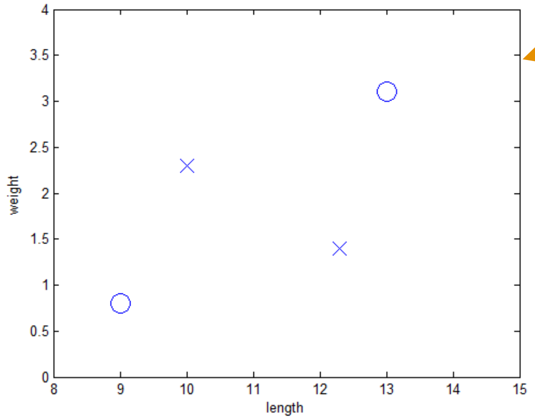


Test our decision stump

- Example:
  - if weight > 1
    - then class label is 0
  - Else
    - then class label is 1
  - This gives one error in training set!



- But gives three errors in this (strange) test set!



test_set.txt - Bloco de n...

Arquivo Editar Formatar Exibir Ajuda

13	3.1	0
9	0.8	0
12.3	1.4	1
10	2.3	1

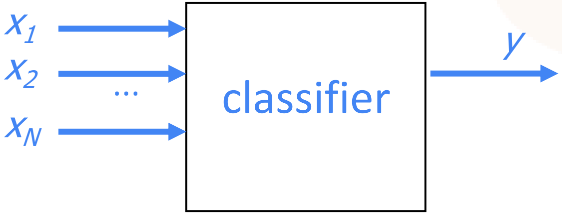
Classification and regression problems

- Both rely on **supervised learning**: when training the model, we know the correct **output** *y*
- The **input** is a vector $x=[x_1, \ldots, x_N]$ with *N* **features**

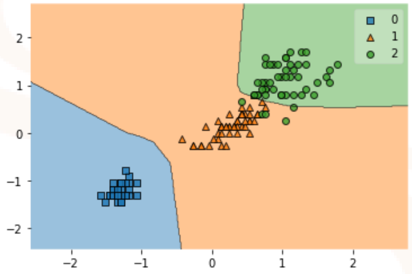
Differences:

Classification

- Output *y* is an element of a set {1,…, *Y*} of *Y* labels.
- Evaluation is based e.g. on misclassification (or error) rate

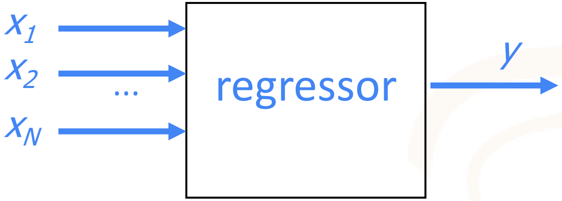


Classification example: input $[x_1, x_2]$, N=2 and $y \in \{0, 1, 2\}$

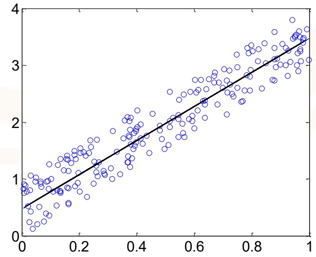


Regression

- Output *y* is a real number or vector (multivariate regression)
- Evaluation is based e.g. on the mean-squared error (MSE), mean absolute error (MAE), etc.

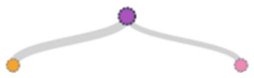


Regression example: input x_1, N=1 and $y \in R$

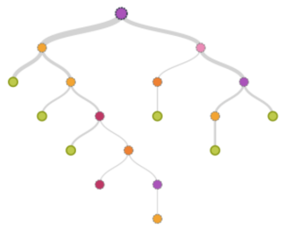


Learning algorithms (most support classification and regression)

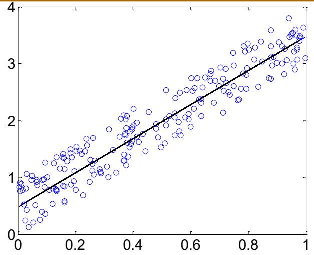
Decision stump (single if/else rule)



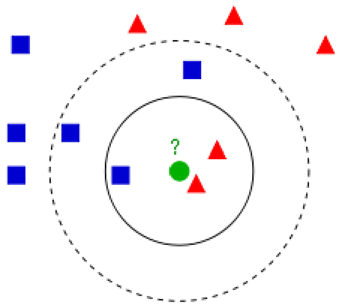
Decision tree



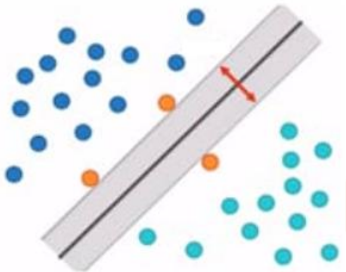
Linear regression



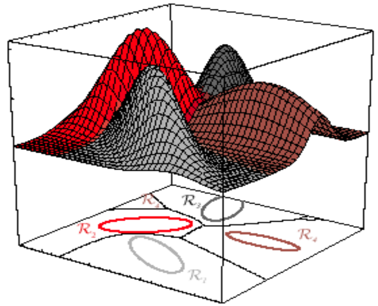
K-nearest neighbors (KNN)



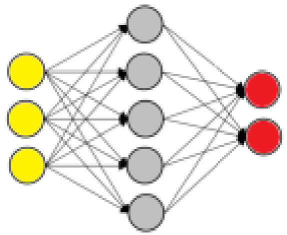
Support vector machine (SVM)



Naïve Bayes

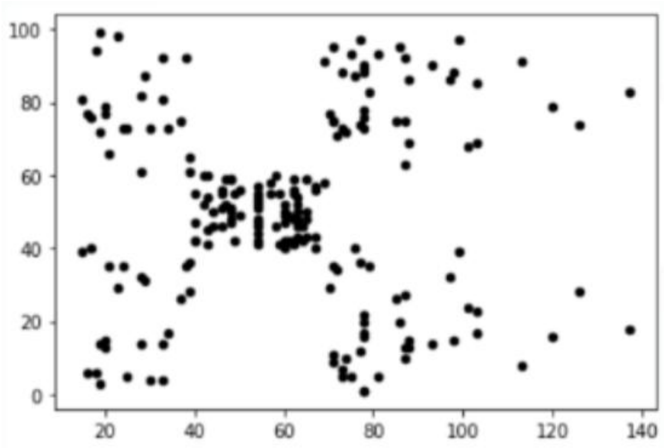


Artificial neural network (ANN)

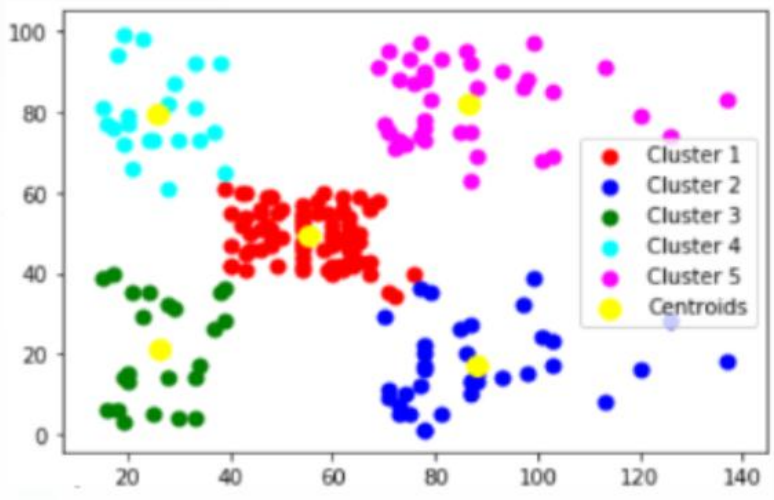


Alternative to supervised learning: unsupervised

Unsupervised learning



Unlabeled data

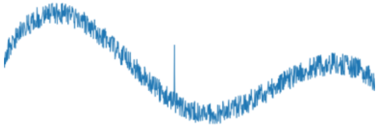


K-means clustering with K=5 centroids

Adapted from https://ai.plainenglish.io/what-is-k-means-clustering-3060791cb589?gi=ba862c73fd17

Popular special case of unsupervised learning: anomaly detection

Examples of anomalies (univariate time series)



Spike

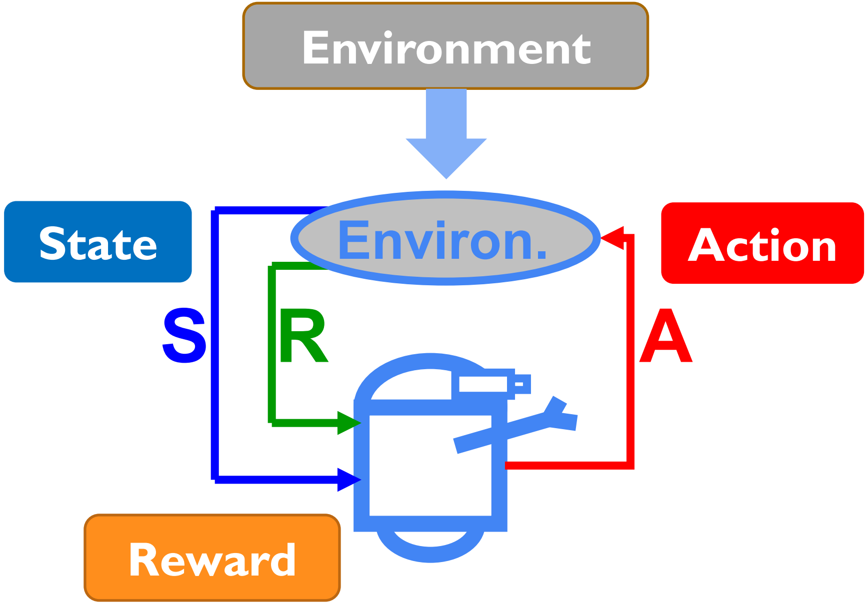
Level shift

Pattern change

Anomaly in seasonal patterns

[1] Anomaly Detection Toolkit (ADTK) - https://pypi.org/project/adtk/

Distinct from supervised and unsupervised learning: Reinforcement Learning (RL)



Online learning, no need for output labels. Support to delayed reward

Goal: Find a policy that maximizes the ***return*** over a lifetime (episode, if not a continuing task), not the immediate ***reward***

Deep learning frameworks

DL training frameworks implement sophisticated algorithms, which use "backpropagation", automatic differentiation and stochastic gradient descent (SGD)

python — Most used language

PyTorch — Facebook's / Meta's

TensorFlow & Keras — Google's, TF versions 1 and 2, with high level Keras API

Deployment frameworks: facilitate pruning the models and quantizing the weights for acceleration

Qualcomm's AI Model Efficiency Toolkit

www.tinyml.org

TorchScript, Tensorflow Lite & PyTorch Quantization

Other tools: NVIDIA, Intel, etc.

Auxiliary tools for (shallow) machine learning, debugging, assessing models and running on cloud

Scikit learn

jupyter

It may not be trivial to set up your development workflow